Tracing the Geometry of Compositional Generalization in a Diffusion U-Net

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Abstract

Diffusion models can perform compositional generalization, combining familiar features in novel ways, but how this ability is acquired internally remains unclear. We study the transition from memorization to compositional generalization in a conditional U-Net trained on a diffusion task with images of three binary factors. Representational changes are concentrated in the decoder, where feature geometry reorganizes and representation dimensionality expands, while encoder representations remain comparatively stable. Causally, transplanting post-generalization decoder blocks Up1+Up2 weights into the pre-generalization model recovers 93% of the out-of-distribution performance gap. Steering learned feature directions also selectively controls generated attributes. These results identify localized decoder changes associated with the emergence of compositional generalization, advancing our understanding of how generative models acquire generalizing behavior.

Keywords: Compositional generalization; diffusion models; representation geometry; mechanistic interpretability; U-Net; causal interventions; feature steering; weight patching.

1. Introduction

Compositional generalization—the ability to recombine familiar factors into unseen combinations—is an important capability of generative models. Conditional diffusion models can acquire this ability during training, transitioning from memorizing observed combinations to generating increasingly distant out-of-distribution (OOD) compositions (Okawa et al., 2025). Prior work has linked successful compositionality to factorized and approximately orthogonal representations (Liang et al., 2024; Uselis et al., 2026), while also showing that factorization at an intermediate representation is insufficient if downstream layers re-entangle the factors (Liang et al., 2025). However, it remains unclear where within a diffusion model the representations change as compositional generalization is acquired, and whether those changes are causally responsible for the new behavior.

We investigate this question in the conditional U-Net setting of Okawa et al. (2025), where images are defined by three binary factors—shape, colour and size—and half of their combinations are held out during training. We track feature geometry across U-Net blocks and use causal interventions to localize the transition.

2. Related work

Okawa et al. (2025) introduced the discrete compositional task used here and characterized its sequential transition from memorization to OOD generalization, without analysing the underlying representations. Most closely related, Liang et al. (2024) showed that diffusion models trained on continuous Gaussian factors develop factorized, approximately

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orthogonal manifold representations as compositionality emerges. Their analysis focuses on manifold formation at a selected internal layer; in contrast, we study discrete factors across U-Net layers and causally localize the generalization transition through weight patching and feature steering. Uselis et al. (2026) formalize geometric requirements for compositional generalization with linear readouts, showing that representations must decompose additively into orthogonal concept-specific components. Liang et al. (2025) show that such factorization at an intermediate representation is nevertheless insufficient for generative models if downstream layers re-entangle the factors. Other work characterizes hierarchical representations (Favero et al., 2025), local conditional-score mechanisms (Bradley, 2026), and causal feature control in diffusion models (Tinaz et al., 2025).

3. Methods

We re-trained the conditional U-Net diffusion model of Okawa et al. (2025). Images vary along three binary factors—shape (circle/triangle), size (large/small), and colour (blue/red)—yielding eight combinations. The model is trained on four combinations (000, 100, 010, 001) and evaluated on four held-out combinations: three at Hamming distance 1 (110, 101, 011) and one at distance 2 (111; Figure S1). Code: https://anonymous.4open.science/r/geometry_compgen_diffusion-DAB6/.

4. Results

We reproduce the sequential compositional-generalization pattern of Okawa et al. (2025): the model memorizes the training combinations by step 1,000 and generalizes to held-out combinations by step 5,000, with colour learned last (Figure S2). We compare internal representations immediately before and after this transition.

4.1. Representational changes are concentrated in the decoder

We first compared feature-vector geometry across U-Net blocks at steps 1,000 and 5,000 (Figure 1). Encoder and bottleneck representations changed little, whereas the largest changes occurred at Up1 and Up2. At Up1, colour becomes approximately orthogonal to shape and size while shape and size remain partially aligned. Up2 undergoes a different reorganization, with shape becoming strongly aligned with both colour and size. Feature directions were largely invariant to context at both checkpoints (most context-conditional cosine similarities > 0.85; Figure S4), suggesting substantial factorization before compositional generalization emerged.

Feature representations remained low-dimensional throughout training. Shape was already two-dimensional at Up0 and Up2, while colour and size remained predominantly one-dimensional. The only clear dimensionality change across the generalization transition occurred at Up1, where shape expanded from a one-dimensional direction to a two-dimensional subspace (Table S1). Together, these results localize the main representational reorganization accompanying generalization to the decoder.

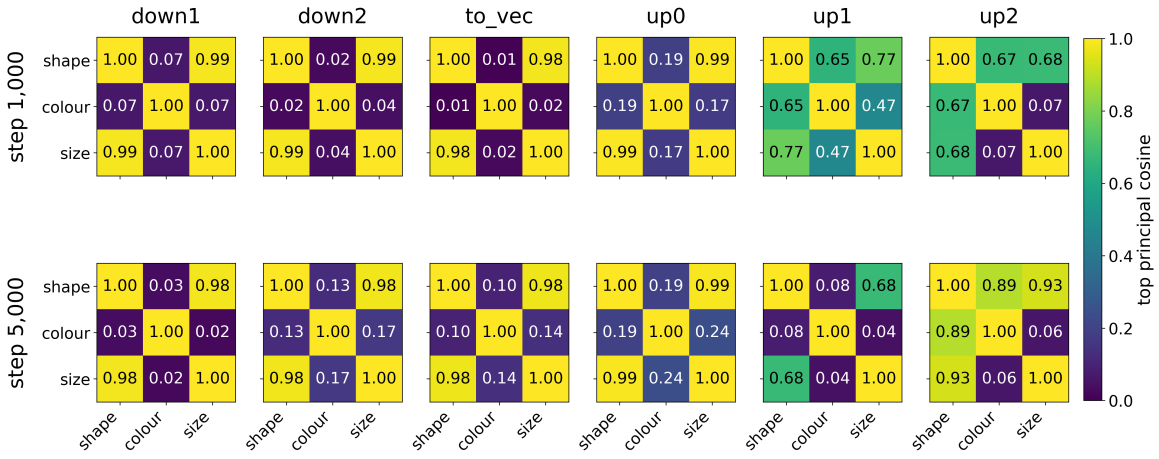


Figure 1: Feature-vector geometry before and after compositional generalization. Pairwise cosine similarity between shape, colour, and size directions at steps 1,000 and 5,000. Geometry is comparatively stable in the encoder, with the largest changes in Up1 and Up2

4.2. Post-generalization decoder changes causally support compositional generalization

Cross-checkpoint weight patching further localized the transition (Figure 2). Replacing Up1+Up2 in the pre-generalization model with their post-generalization weights increased OOD exact-combination accuracy by 0.680, recovering 93% of the pre-to-post generalization gap. In contrast, patching Up0 produced only a small improvement (+0.045), and adding Up0 to Up1+Up2 provided no detectable additional benefit (Table S3). Thus, the learned changes sufficient to recover most OOD generalization are concentrated in the conditioned Up1 and Up2 blocks.

The feature directions in these layers were also functionally meaningful (Figure 2). Steering shape, colour, or size produced strong effects on the corresponding generated attribute relative to magnitude-matched random directions (on-target $\Delta b = 0.508$–0.599; all $q \approx 10^{-4}$), with generally high relative specificity ($S_{\text{rel}} = 0.773$ – 0.937 ; Table S5). This shows that the directions identified by the representational analysis can selectively control generation.

Finally, directly restoring pre-generalization colour alignment at Up1/Up2 altered generated representations, but the effect on the primary colour-projection metric was not exceptional relative to matched random rotations with alignment held fixed. Secondary off-axis effects were stronger (Table S6).

5. Conclusions

Our results complement Liang et al. (2024), who showed that diffusion models develop factorized manifold representations of continuous features as compositionality emerges. Here,

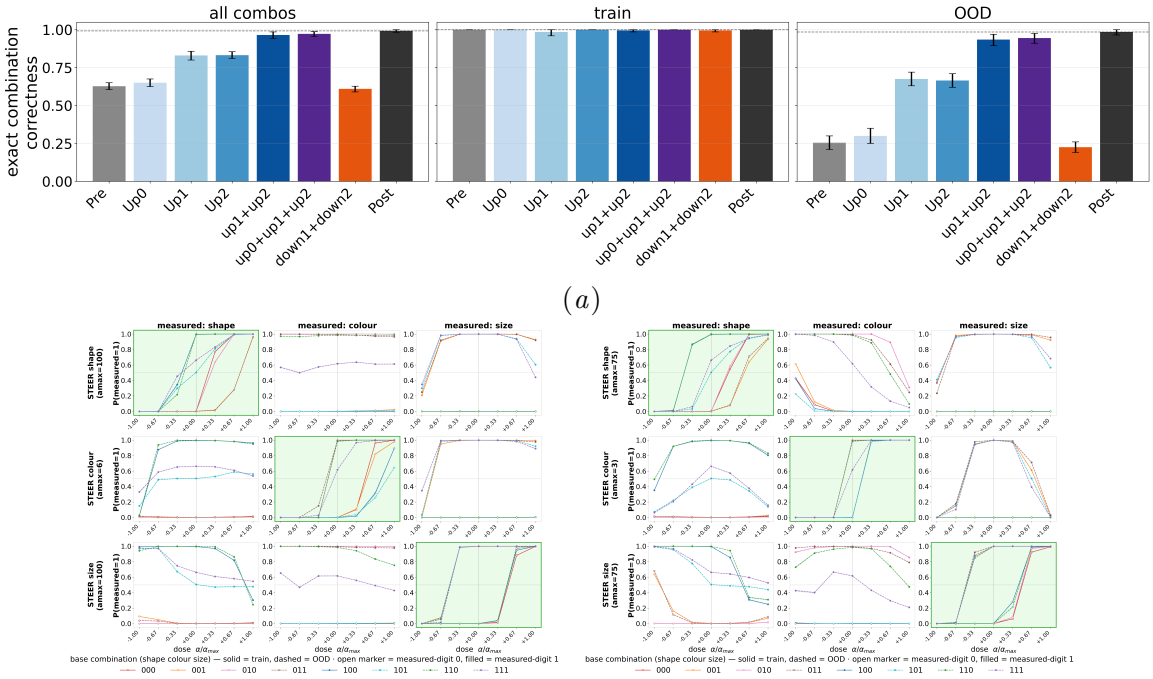


Figure 2: Intervention experiments. (A) Cross-checkpoint weight patching: replacing pre-generalization blocks with post-generalization weights shows that Up1+Up2 recover 93% of the OOD performance gap, with little additional contribution from Up0. Bars show changes in exact-combination accuracy with 95% bootstrap CIs. (B-C) Feature-vector steering at Up1 and Up2 selectively controls shape, colour, and size across normalized steering strengths.

using discrete categorical features, we find that this transition is not uniform across the network: representations remain comparatively stable in the encoder while feature geometry reorganizes in conditioned decoder blocks. Critically, post-generalization Up1+Up2 weights recover most of the OOD performance gap when transplanted into the pre-generalization model, and their feature directions support selective causal steering. This suggests that compositional generalization can arise through localized changes in how downstream layers represent and render already-learned factors, rather than requiring a network-wide representational transition.

5.1. Limitations

The study focuses on a single small U-Net, so the generality of the mechanism remains unknown. Current analyses focuses on a fixed denoising timestep ($t = 200$). Finally, weight patching establishes sufficiency of the decoder changes, but it does not establish that the observed geometric changes themselves are necessary for generalization.

Acknowledgments

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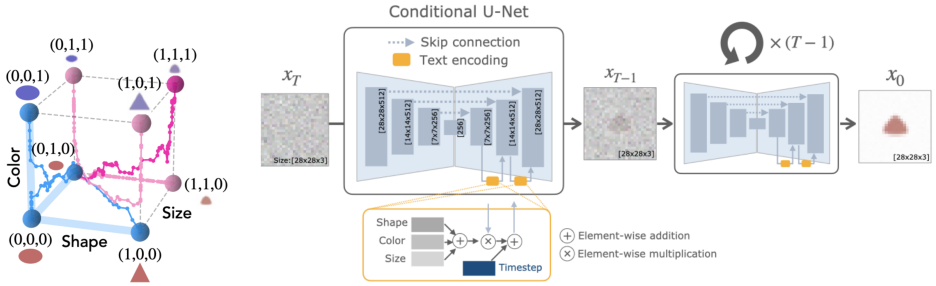
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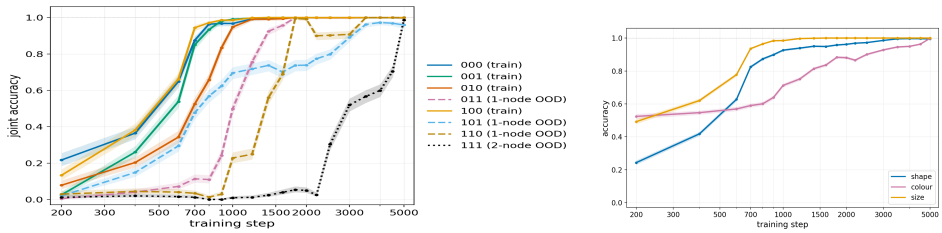
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Appendix A. Supplementary material



(a) (b)

Figure S1: Model architecture and compositional task. (A) The eight combinations of three binary factors (shape, colour, size); four are used for training (blue), while four are held out at one (light pink) or two (dark pink) nodes in Hamming distance. (B) Conditional diffusion model architecture, adapted from Okawa et al. (2025). The denoising process is parameterized by a U-Net with downsampling and upsampling blocks connected by skip connections and conditioned on the target feature classes. Reproduced from Okawa et al. (2025)



(a) (b)

Figure S2: Emergence of compositional generalization during training. (A) Exact-combination accuracy for training and held-out combinations. The model memorizes the training set by step 1,000 and generalizes to all held-out combinations by step 5,000. (B) Feature-wise accuracy across training. Shape and size are learned earlier, while colour is the last feature to reach high accuracy.

A.0.1. Steering results and controls

On-target steering was substantially stronger than matched random directions ($\Delta b = 0.508$–0.599; all $q \approx 10^{-4}$). The largest off-target effects were Up2 shape→colour (-0.265) and Up2 size→shape (-0.216).

Shape and size steering were less selective at Up2 than Up1, whereas colour was more selective at Up2.

A.0.2. Colour-vector rotation results

Restoring pre-generalization colour alignment reduced y_{rb} at both Up1 and Up2, with the largest and approximately additive effect when both layers were rotated jointly (Table S6; all $p \approx 10^{-4}$).

The joint y_{rb} effect remained within the distribution of $N = 50$ matched random rotations (16th percentile overall; 8th OOD). The off-axis residual was more extreme than all controls for blue ($z = -2.69$) and red ($z = 3.60$). Historical and control perturbation magnitudes differed by less than 0.45% at $s = 1$.

Appendix B. Additional methods

B.1. Diffusion model and dataset

Dataset and train–test split. We use the compositional dataset of Okawa et al. (2025), consisting of images defined by three binary factors: shape, colour, and size. Of the eight possible combinations, four (000, 100, 010, 001) are used for training. The remaining combinations are held out: 101, 011, and 110 are one node from the training set in Hamming distance, while 111 is two nodes away.

Conditional diffusion model. We use the publicly available conditional U-Net implementation of Okawa et al. (2025). The architecture contains two downsampling blocks (`down1`, `down2`), a bottleneck projection (`to_vec`) that reduces the representation to 1×1,

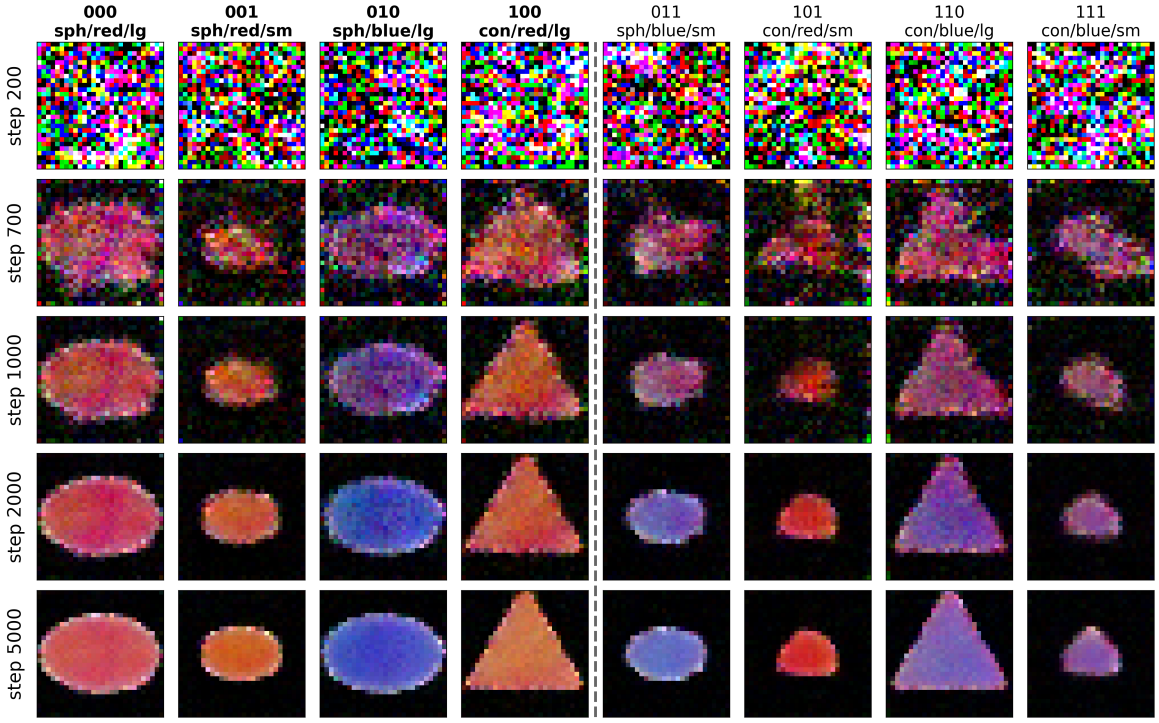


Figure S3: Generated examples throughout training. Example generations for each feature combination across training checkpoints. Samples progressively improve in fidelity to the conditioning features, from poor or incomplete compositions early in training to accurate generation of both training and held-out combinations at later checkpoints.

an initial decoder expansion (`up0`) that restores it to 7×7, and two conditioned upsampling blocks (`up1`, `up2`). Conditioning labels are embedded and injected into the upsampling blocks `up1` and `up2`. Training follows Okawa et al. (2025) using Adam ($\beta_1 = 0.9$, $\beta_2 = 0.99$), weight decay 10^{-5}, and $T = 400$ diffusion steps.

Evaluation. Following Okawa et al. (2025), separate linear classifiers are trained to predict shape, colour, and size. Exact-combination accuracy requires all three features to be classified correctly:

$$\text{Accuracy} = \frac{1}{N_t} \sum_{n=1}^{N_t} \prod_{f=1}^{3} \mathbf{1}\left[\hat{v}_f^{(n)} = v_f^{(n)}\right], \tag{1}$$

where $\hat{v}_f^{(n)}$ and $v_f^{(n)}$ are the predicted and target values of feature f for sample n.

B.2. Feature representation analysis

Activation extraction. We extract representations from `down1`, `down2`, `to_vec`, `up0`, `up1`, and `up2` at checkpoints spanning the memorisation–generalisation transition. For

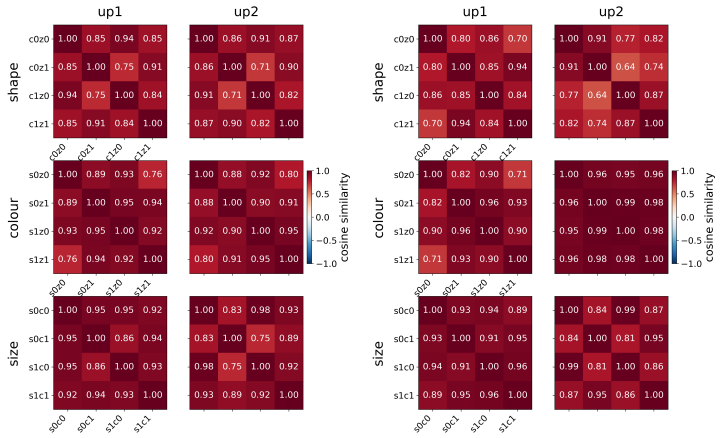


Figure S4: Context dependence of feature directions in Up1 and Up2. Pairwise cosine similarity between context-conditional feature vectors at step 1,000 (A) and step 5,000 (B). Feature representations are highly consistent across contexts at both checkpoints (most cosine similarities >0.85; all >0.7 except one at ~ 0.64)

each composition, $N = 200$ labelled images are corrupted to a fixed diffusion timestep $t = T/2 = 200$ using fixed noise seeds. Ground-truth conditioning labels are supplied during the forward pass, and module outputs are recorded using forward hooks.

For activations $A^{(b)} \in \mathbb{R}^{N\times C\times H\times W}$, each sample is reduced to a channel vector by global spatial average pooling:

$$h_i^{(b)} = \frac{1}{HW}\sum_{u=1}^{H}\sum_{v=1}^{W} A^{(b)}_{i,:,u,v} \in \mathbb{R}^C. \tag{2}$$

Each composition b is represented by its centroid:

$$\mu(b) = \frac{1}{N}\sum_{i=1}^{N} h_i^{(b)} \in \mathbb{R}^C. \tag{3}$$

The same timestep and noise protocol is used across checkpoints.

Global feature vectors and geometry. For feature f, the global feature direction is defined as the difference between centroid means for its two values, marginalising over the other factors:

$$v_f = \operatorname*{mean}_{b:\, b_f=1} \mu(b) - \operatorname*{mean}_{b:\, b_f=0} \mu(b). \tag{4}$$

Cross-feature geometry is measured using pairwise cosine similarity:

$$\cos(v_f, v_g) = \frac{v_f^\top v_g}{\|v_f\|\,\|v_g\|}. \tag{5}$$

Table S1: Bootstrapped effective rank across layers and features at training steps 1,000 and 5,000. $\hat{k}$ denotes the effective rank required to explain 90% of the energy, $\pi(k = 2)$ is the bootstrap probability of obtaining rank 2, and ef_1 is the fraction of energy explained by the first singular direction. Results use $B = 10{,}000$ bootstrap resamples.

		Step 1,000			Step 5,000			Change		
Layer	Feature	$\hat{k}_{90}$	$\pi(k=2)$	ef_1	$\hat{k}_{90}$	$\pi(k=2)$	ef_1	$\Delta\mathrm{ef}_1$	$\Delta\pi$	k_{95}
Down1	Shape	1	0.000	0.984	1	0.000	0.987	+0.003	0.000	$1 \to 1$
Down1	Colour	1	0.000	0.981	1	0.000	0.986	+0.005	−0.000	$1 \to 1$
Down1	Size	1	0.000	0.993	1	0.000	0.990	−0.003	0.000	$1 \to 1$
Down2	Shape	1	0.000	0.988	1	0.000	0.990	+0.002	0.000	$1 \to 1$
Down2	Colour	1	0.095	0.969	1	0.103	0.972	+0.003	+0.008	$1 \to 1$
Down2	Size	1	0.000	0.996	1	0.000	0.997	+0.000	0.000	$1 \to 1$
ToVec	Shape	1	0.000	0.971	1	0.000	0.972	+0.001	0.000	$1 \to 1$
ToVec	Colour	1	0.098	0.965	1	0.099	0.968	+0.003	+0.000	$1 \to 1$
ToVec	Size	1	0.000	0.993	1	0.000	0.993	−0.000	0.000	$1 \to 1$
Up0	Shape	2	1.000	0.786	2	1.000	0.740	−0.046	0.000	$2 \to 2$
Up0	Colour	1	0.590	0.933	1	0.334	0.943	+0.013	−0.256	$2 \to 2$
Up0	Size	1	0.000	0.978	1	0.000	0.972	−0.006	0.000	$1 \to 1$
Up1	Shape	1	0.050	0.908	2	1.000	0.888	−0.020	+0.950	$2 \to 2$
Up1	Colour	1	0.000	0.932	1	0.265	0.907	−0.024	+0.265	$2 \to 2$
Up1	Size	1	0.000	0.948	1	0.000	0.952	+0.002	0.000	$2 \to 2$
Up2	Shape	2	0.914	0.888	2	1.000	0.844	−0.044	+0.086	$2 \to 2$
Up2	Colour	1	0.000	0.936	1	0.000	0.974	+0.039	0.000	$2 \to 1$
Up2	Size	1	0.017	0.911	1	0.000	0.913	+0.002	−0.017	$2 \to 2$

Table S2: Mean slope [95% CI] for each measured feature (columns) across feature vector steering experiments (rows) in up1.

	Shape	Colour	Size
Steer: shape	+0.544 [0.543, 0.545]	+0.007 [0.006, 0.007]	+0.074 [0.072, 0.075]
Steer: colour	+0.108 [0.106, 0.110]	+0.526 [0.525, 0.528]	+0.137 [0.136, 0.139]
Steer: size	−0.136 [−0.137, −0.134]	−0.026 [−0.026, −0.025]	+0.526 [0.526, 0.526]

Context-conditional feature vectors. To measure whether a feature direction depends on the other factors, we compute feature vectors while holding the other two features fixed:

$$v_f(\kappa) = \mu(b_f = 1, \kappa) - \mu(b_f = 0, \kappa), \qquad \kappa = 1, \ldots, 4, \tag{6}$$

where κ indexes the four possible contexts defined by the remaining two binary factors.

Table S3: Cross-checkpoint weight-patching results. Effects are changes in exact-combination accuracy relative to the pre-generalization model. Gap closed denotes the fraction of the pre- to post-generalization OOD performance difference recovered by the intervention.

Intervention	Δ accuracy	95% CI	p
Up1+Up2 (OOD)	+0.680	[0.615, 0.740]	$\approx 10^{-4}$
Up0 (OOD)	+0.045	[0.010, 0.080]	0.019
Up1+Up2 vs. Up0	+0.635	[0.565, 0.700]	$\approx 10^{-4}$
Up0 added to Up1+Up2	+0.010	[−0.015, +0.035]	0.69

Up1+Up2 recovered 93% of the pre-to-post OOD performance gap.

Table S4: Difference in steering slope between learned and magnitude-matched random directions. Rows indicate the steered feature and columns the measured feature. All effects were significant after BH–FDR correction ($q \approx 10^{-4}$).

Layer	Steered feature	Shape	Colour	Size
Up1	Shape	+0.545	+0.011	+0.074
Up1	Colour	+0.106	+0.508	+0.135
Up1	Size	−0.138	−0.026	+0.526
Up2	Shape	+0.548	−0.265	+0.071
Up2	Colour	+0.056	+0.599	+0.057
Up2	Size	−0.216	−0.041	+0.545

Feature dimensionality. For each feature, the four context-conditional vectors are stacked into $M_f \in \mathbb{R}^{4\times C}$ and decomposed using an uncentered singular value decomposition:

$$M_f = U\Sigma V^\top. \tag{7}$$

The rows are not mean-centred so that the global feature direction is retained in the leading component. Effective dimensionality is defined as the minimum number of components explaining 90% of the squared singular-value energy:

$$k_{90} = \min\left\{ k : \frac{\sum_{j=1}^{k}\sigma_j^2}{\sum_j \sigma_j^2} \geq 0.90 \right\}. \tag{8}$$

Thus, $k_{90} = 1$ indicates an approximately context-independent direction, whereas $k_{90} > 1$ indicates that the feature spans multiple directions across contexts.

B.3. Causal interventions

Interventions are applied to module outputs using forward hooks and remain active throughout the reverse diffusion process. Initial noise and conditioning labels are fixed across

Table S5: Steering specificity across layers and features.

Layer	Feature	S_f	S_{rel}
Up1	Shape	0.504	0.935
Up1	Colour	0.404	0.805
Up1	Size	0.445	0.878
Up2	Shape	0.381	0.773
Up2	Colour	0.592	0.937
Up2	Size	0.414	0.809

Table S6: Slope of signed correct-colour projection (y_{rb}) per unit rotation strength. d_{geom} denotes the joint Up1+Up2 effect minus the matched control.

Rotation	All	Train	OOD	Red	Blue
Up1	−0.0040	−0.0026	−0.0055	−0.0011	−0.0070
Up2	−0.0063	−0.0071	−0.0055	−0.0058	−0.0068
Up1+Up2	−0.0103	−0.0096	−0.0109	−0.0069	−0.0136
Control	−0.0045	−0.0040	−0.0050	−0.0030	−0.0059
d_{geom}	−0.0058	−0.0057	−0.0059	−0.0039	−0.0077

intervention conditions, and generated images are evaluated using the feature classifiers described above.

Weight patching. To localise learned parameter changes across training, selected modules in the pre-generalisation model are replaced with the corresponding parameters and buffers from the post-generalisation checkpoint. Encoder blocks, `up0`, conditioned decoder blocks, time embeddings, and conditioning embeddings are patched individually or in groups; the parameter-free `to_vec` projection is omitted. The patched model then generates normally using its own activations.

Feature-vector steering. For the unit feature direction $\hat{d}_f = d_f/\|d_f\|$, steering modifies the target-layer activation at strength α:

$$h'_{n,c,u,v} = h_{n,c,u,v} + \alpha \hat{d}_{f,c}. \tag{9}$$

Steering strength is varied around zero, with $\alpha = 0$ corresponding to unperturbed generation.

To control for generic perturbation effects, each learned direction is compared with $R = 10$ magnitude-matched isotropic random directions. For each trajectory, we fit the slope of probe probability $P(f = 1)$ against normalized steering dose $\alpha/\alpha_{\max}$ and compute

$$\Delta b = b_{\text{real}} - \frac{1}{R}\sum_{r=1}^{R} b_{\text{random},r}. \tag{10}$$

Statistics use $n = 200$ combo-averaged blocks, bootstrap confidence intervals, two-sided sign-flip tests, and BH–FDR correction.

Steering specificity is summarized using

$$S_f = b_{f\to f} - \frac{|b_{f\to f_1}| + |b_{f\to f_2}|}{2}, \tag{11}$$

and the scale-free score

$$S_{\text{rel}} = 1 - \frac{\text{mean}\,|\beta_{\text{off}}|}{|\beta_{\text{target}}|}. \tag{12}$$

Larger values indicate greater feature specificity; $S_{\text{rel}} = 1$ indicates no off-target effect. Steering slopes are reported per normalized dose; $\alpha_{\max}$ was 100/6/100 for shape/colour/size at Up1 and 75/3/75 at Up2.

Vector rotation. To modify feature geometry while preserving the activation coefficient along the feature direction, we replace the component along $\hat{c}$ with the same coefficient along a rotated direction:

$$\begin{aligned} h' &= h - a\hat{c} + a\hat{c}_\theta, \\ a &= h^\top \hat{c}, \\ \hat{c}_\theta &= \cos(\theta)\hat{c} + \sin(\theta)\hat{s}_\perp, \end{aligned}$$

where $\hat{s}_\perp$ is orthogonal to $\hat{c}$.

For colour, the post-generalization direction was rotated toward its pre-generalization alignment at Up1, Up2, or both layers jointly; $s = 1$ restores the historical alignment. We compared these interventions with magnitude-matched constant-alignment controls and with $N = 50$ matched random rotations. Geometry-specific effects were quantified as

$$d_{\text{geom}} = b_{\text{Up1+Up2}} - b_{\text{control}},$$

where b is the slope of signed correct-colour projection (y_{rb}) against rotation strength. We additionally measured the off-axis colour residual (resid_{rb}).